

PRACTICAL WEEK 3 (DBMS)

TASK 1: Document Modeling — create a book_metadata collection with embedded patterns (reviews list inside the book document) and referenced patterns (linking to member_id in SQL). 2) Advanced Querying — use comparison operators (\$gt, \$in) to find books rated above 4, and a regex query to find books with "Digital" formats. 3) Aggregation Pipeline — match books published after 2020, \$unwind the reviews array, then \$group by title and compute the average rating. 4) Indexing for Speed — create a text index on the review comments field and use .explain() to check whether the query uses the index.

The screenshot shows the MongoDB Data Explorer interface. The left sidebar displays the 'Data Explorer' section with a tree view of clusters. The main panel shows the 'book_metadata' collection. The 'Documents' tab is active, displaying a list of documents. The first document is for 'The Martian' by Andy Weir, published in 2021, with a digital PDF format and 3 reviews. The second document is for 'Dune' by Frank Herbert, published in 2022, with a digital EPUB format and 3 reviews. The third document is for 'Clean Code' by Robert C. Martin, published in 2019, with a physical format and 2 reviews. The fourth document is for 'Artificial Intelligence Basics' by John Smith, published in 2023, with a digital PDF format and 3 reviews. The fifth document is for 'Data Structures' by an unknown author, published in 2023, with a physical format and 3 reviews.

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookflow
 - book_metadata**
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1 - 5 of 5

```
{
  "_id": ObjectId('6a963e0c7b9df3ab7aeeaa0'),
  "book_id": 184,
  "title": "Artificial Intelligence Basics",
  "author": "John Smith",
  "published_year": 2023,
  "category": "Artificial Intelligence",
  "format": "Digital PDF",
  "file_size": "4.2 MB",
  "member_id": 184,
  "reviews": Array (3)
}
```

```
{
  "_id": ObjectId('6a963efcc7b9df3ab7aeeaa1'),
  "book_id": 185,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "5.0 MB",
  "member_id": 185,
  "reviews": Array (2)
}
```

Alert System Status: All Good

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TASK 2: Advanced Querying

Query 1:

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Data Explorer

My Queries Data Modeling

CLUSTERS (1)

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Documents 5 Aggregations Schema Indexes 1 Validation

[Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1 - 5 of 5

```
{
  "reviews.rating": {
    "$gt": 4
  }
}
```

```
{
  "_id": ObjectId('6a963e74c7b9df3ab7aeeaa2'),
  "book_id": 181,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 181,
  "reviews": Array (3)
}
```

```
{
  "_id": ObjectId('6a963e7c7b9df3ab7aeeaa4'),
  "book_id": 182,
  "title": "Dune",
  "author": "Frank Herbert",
  "published_year": 2022,
  "category": "Science Fiction",
  "format": "Digital EPUB",
  "file_size": "3.1 MB",
  "member_id": 182,
  "reviews": Array (3)
}
```

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Data Explorer

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Documents 5 Aggregations Schema Indexes 1 Validation

[Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1 - 5 of 5

```
{
  "reviews.rating": {
    "$gt": 4
  }
}
```

```
{
  "_id": ObjectId('6a963ec7c7b9df3ab7aeeaa6'),
  "book_id": 183,
  "title": "Clean Code",
  "author": "Robert C. Martin",
  "published_year": 2019,
  "category": "Programming",
  "format": "Physical",
  "dimensions": Object (3)
  "member_id": 183,
  "reviews": Array (2)
}
```

```
{
  "_id": ObjectId('6a963ed6c7b9df3ab7aeeaa8'),
  "book_id": 184,
  "title": "Artificial Intelligence Basics",
  "author": "John Smith",
  "published_year": 2023,
  "category": "Artificial Intelligence",
  "format": "Digital PDF",
  "file_size": "4.2 MB",
  "member_id": 184,
  "reviews": Array (3)
}
```

Alert System Status: All Good

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Data Explorer

My Queries

Data Modeling

CLUSTERS {}

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- Cluster0
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Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

```
{
  "reviews.rating": {
    "$gt": 4
  }
}
```

```
{
  "_id": ObjectId("6a963efcc7b9df3ab7aee5a5a"),
  "book_id": 184,
  "title": "Artificial Intelligence",
  "format": "Digital PDF",
  "file_size": "4.2 MB",
  "member_id": 184,
  "reviews": Array (3)
}
```

```
{
  "_id": ObjectId("6a963efcc7b9df3ab7aee5a5a"),
  "book_id": 185,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "75.8 MB",
  "member_id": 185,
  "reviews": Array (2)
}
```

Query 2:

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Data Explorer

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Documents 5 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

```
{
  "_id": ObjectId("6a963efcc7b9df3ab7aee5a62"),
  "book_id": 181,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 181,
  "reviews": Array (3)
}
```

```
{
  "_id": ObjectId("6a963efcc7b9df3ab7aee5a64"),
  "book_id": 182,
  "title": "Dune",
  "author": "Frank Herbert",
  "published_year": 2022,
  "category": "Science Fiction",
  "format": "Digital EPUB",
  "file_size": "73.1 MB",
  "member_id": 182,
  "reviews": Array (2)
}
```

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Data Explorer

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Documents 5 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

```
{
  "_id": ObjectId("6a963efcc7b9df3ab7aee5a66"),
  "book_id": 183,
  "title": "Clean Code",
  "author": "Robert C. Martin",
  "published_year": 2019,
  "category": "Programming",
  "format": "Physical",
  "dimensions": Object (3),
  "member_id": 183,
  "reviews": Array (2)
}
```

```
{
  "_id": ObjectId("6a963efcc7b9df3ab7aee5a68"),
  "book_id": 184,
  "title": "Artificial Intelligence basics",
  "author": "John Sedley",
  "published_year": 2023,
  "category": "Artificial Intelligence",
  "format": "Digital PDF",
  "file_size": "4.2 MB",
  "member_id": 184,
  "reviews": Array (3)
}
```

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Data Explorer

My Queries Data Modeling

CLUSTERS {}

Search clusters

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Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

```
{
  "_id": ObjectId("6a963efcc7b9df3ab7aeaa5a"),
  "book_id": 185,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "4.2 MB",
  "member_id": 104,
  "reviews": Array (3)
}
```

```
{
  "_id": ObjectId("6a963efcc7b9df3ab7aeaa5a"),
  "book_id": 185,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "4.2 MB",
  "member_id": 104,
  "reviews": Array (3)
}
```

Query 3:

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Data Explorer

My Queries Data Modeling

CLUSTERS {}

Search clusters

- Cluster0
 - admin
 - bookflow
 - book_metadata
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

```
{
  "_id": ObjectId("6a963e74c7b9df3ab7aeaa62"),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": Array (3)
}
```

```
{
  "_id": ObjectId("6a963eb7c7b9df3ab7aeaa64"),
  "book_id": 102,
  "title": "Dune",
  "author": "Frank Herbert",
  "published_year": 2022,
  "category": "Science Fiction",
  "format": "Digital EPUB",
  "file_size": "3.1 MB",
  "member_id": 102,
  "reviews": Array (3)
}
```

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Project 0

Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

format: {

\$regex: "Digital",

\$options: "i"

ADD DATA

BULK

EXPORT CODE

_id: ObjectId('6a963ed6c7b9df3ab7aea68')

book_id: 184

title: "Artificial Intelligence Basics"

author: "John Smith"

published_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file_size: "4.2 MB"

member_id: 184

reviews: Array (3)

_id: ObjectId('6a963efcc7b9df3ab7aea6a')

book_id: 185

title: "Data Structures"

author: "Mark Allen"

published_year: 2024

category: "Computer Science"

format: "Digital PDF"

file_size: "5.0 MB"

member_id: 185

reviews: Array (2)

TASK 3: Aggregation Pipeline

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PROJECT

Project 0

Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

View monitoring

Visualize Your Data

\$match

Generate aggregation

Explain

Run

Options

Untitled - modified

SAVE

CREATE NEW

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Preview of documents

Stage 1 | \$match

1 {

2 "published_year": {

3 "gte": 2020

4 }

5 }

Output preview after \$match stage (Sample of 4 documents)

_id: ObjectId('6a963e74c7b9df3ab7aea62')

book_id: 181

title: "The Martian"

author: "Andy Weir"

published_year: 2011

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 181

_id: ObjectId('6a963eb7c7b9df3ab7aea64')

book_id: 182

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 182

_id: ObjectId('6a963ed6c7b9df3ab7aea68')

book_id: 184

title: "Artificial Intelligence Basics"

author: "John Smith"

published_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file_size: "4.2 MB"

member_id: 184

Learn more about aggregation pipeline stages

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Data Explorer

My Queries Data Modeling

CLUSTERS {}

Search clusters

Cluster0

- admin
- bookflow
- book_metadata
- local
- sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate aggregation + Explain Run Options

Preview

Stage 2: \$unwind

Output preview after \$unwind if stage (Sample of 10 documents)

```

1 {
  "_id": "6a963e74c7b9df3ab7aea62",
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2011,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "number": 101
}

```

+ Add stage

Learn more about aggregation pipeline stages

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Data Explorer

My Queries Data Modeling

CLUSTERS {}

Search clusters

Cluster0

- admin
- bookflow
- book_metadata
- local
- sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate aggregation + Explain Run Options

Preview

Stage 3: \$group

Output preview after \$group if stage (Sample of 4 documents)

```

1 {
  "_id": "title",
  "average_rating": {
    "$avg": "$reviews.rating"
  },
  "total_reviews": {
    "$sum": 1
  }
}

```

+ Add stage

Learn more about aggregation pipeline stages

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Data Explorer

My Queries Data Modeling

CLUSTERS {}

Search clusters

Cluster0

- admin
- bookflow
- book_metadata
- local
- sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate aggregation + Explain Run Options

Preview

Stage 4: \$sort

Output preview after \$sort if stage (Sample of 4 documents)

```

1 {
  "average_rating": -1
}

```

+ Add stage

Learn more about aggregation pipeline stages

TASK 4: Indexing for Speed

Create Index

Refresh

Looking for search indexes?
These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR ⓘ	36.9 kB	6 (since Tue Sep 01 2026)	UNIQUE ⓘ	READY
> reviews.comments_text	TEXT ⓘ	20.5 kB	0 (since Tue Sep 01 2026)		READY

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Cluster0

book_metadata

+

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 2

Validation

{

"\$text": {

"\$search": "excellent"

}

}

ADD DATA

BULK

EXPORT CODE

_id: ObjectId('6a963e74c7b9df3ab7aeea62')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

reviews: Array (3)

_id: ObjectId('6a963ec7c7b9df3ab7aeea66')

book_id: 103

title: "Clean Code"

author: "Robert C. Martin"

published_year: 2019

category: "Programming"

format: "Physical"

dimensions: Object (3)

member_id: 103

reviews: Array (2)



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Data Explorer



My Queries



Data Modeling



CLUSTERS (1)



Search clusters



Cluster0



admin



bookflow



book_metadata



local



sample_mflix

Cluster0

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 2

Validation



{

"\$text": {

"\$search": "excellent"

}

}

ADD DATA

BULK

EXPORT CODE

_id: ObjectId('6a963eb7c7b9df3ab7aeea64')

book_id: 102

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 102

reviews: Array (3)

_id: ObjectId('6a963efcc7b9df3ab7aeea6a')

book_id: 105

title: "Data Structures"

author: "Mark Allen"

published_year: 2024

category: "Computer Science"

format: "Digital PDF"

file_size: "5.0 MB"

member_id: 105

reviews: Array (2)



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PROJECT

Project 0 ▾



Data Explorer



My Queries



Data Modeling



CLUSTERS (1)



Cluster0

admin

bookflow

book_metadata

local

sample_mflix

Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 2

Validation

 {
 "\$text": {
 "\$search": "excellent"
 }
}

+

 ADD DATA ▾



 BULK ▾

</>

 EXPORT CODE

published_year : 2024

category : "Computer Science"

format : "Digital PDF"

file_size : "5.0 MB"

member_id : 105

▸ reviews : Array (2)

_id: ObjectId('6a963ed6c7b9df3ab7aeaa68')

book_id : 104

title : "Artificial Intelligence Basics"

author : "John Smith"

published_year : 2023

category : "Artificial Intelligence"

format : "Digital PDF"

file_size : "4.2 MB"

member_id : 104

▸ reviews : Array (3)